

REVIEW ARTICLE

Calcium and hypertension

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A possible relationship between dietary calcium intake and blood pressure has been the focus of considerable recent observational and experimental research. To provide a current perspective, we have reviewed and summarized the epidemiologic data, evidence suggesting abnormalities in calcium metabolism in human hypertension, studies evaluating calcium supplementation in various hypertension populations, and data that may help elucidate possible blood pressure-lowering mechanisms of calcium supplementation. An important relationship between "calcium metabolism" and "salt-sensitivity" is emerging and appears to be providing insights into potential mechanisms that account for the antihypertensive properties of dietary calcium in certain individuals. More experimental work needs to be performed to further define the individuals with hypertension who demonstrate a blood pressure-lowering response to dietary calcium supplementation. (J LAB CLIN MED 1989;114:338-48)

Abbreviations: ATPase = adenosine triphosphatase; PTH = parathyroid hormone; RDA = recommended daily allowance; SHR = spontaneously hypertensive rat

Over the past decade evidence has been accumulated indicating that deficient Ca^{2+} intake and arterial hypertension may be causally related. This evidence has been garnered from epidemiologic reports, from findings of biochemical markers that are consistent with Ca^{2+} deficiency in animal models of hypertension and in patients with hypertension, and from observations of reductions in blood pressure that are associated with dietary Ca^{2+} supplementation. In this review we will examine the evidence for a role of dietary Ca^{2+} intake in the regulation of blood pressure, with an emphasis on epidemiologic, biochemical, hemodynamic, and dietary intervention data in human subjects.

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EPIDEMIOLOGIC EVIDENCE LINKING CALCIUM INTAKE AND BLOOD PRESSURE

A variety of epidemiologic reports have examined the relationship between blood pressure and dietary Ca^{2+} intake. These reports encompass a relatively broad age range, diverse geographic areas, and a variety of ethnic and racial groups. The first National Health and Nutrition Examination Survey performed by the National Center for Health Statistics is one of the largest data bases that has been exploited to evaluate this relationship.^{1,2} Analysis of these data resulted in the interpretation that there was a significant inverse correlation between dietary Ca^{2+} intake and systolic blood pressure that was independent of such variables as age, sex, body mass index, race, and ethnicity.^{1,2} However, as pointed out by Kaplan and Meese,³ there are some concerns about these conclusions. The first National Health and Nutrition Examination Survey used a 24-hour recall method, which is generally acknowledged to be an inadequate recall period.⁴ Excluding people with a history of hypertension may have introduced an uncontrolled variable.³ Finally, it has been pointed out that if age and weight would have been properly incorporated into the statistics used by McCarron et al.,²

the relationship between Ca^{2+} ingestion and blood pressure may have lost its statistical significance.⁵ Nevertheless, the relationship between hypertension and decreased Ca^{2+} intake has been observed in a number of reports containing data obtained from Europe, Puerto Rico, and in both rural and urban environments in the United States.⁶⁻¹⁶

These epidemiologic data also give some indication of the levels of Ca^{2+} intake that may be critical for observation of a relationship between Ca^{2+} intake and blood pressure. From a 4-year prospective study of approximately 60,000 women participating in the Nurses Health Study, a 23% reduction in the risk of developing hypertension was associated with a regular Ca^{2+} intake above 800 mg/day, compared with an intake below 400 mg/day.¹⁷ These data are generally consistent with prior observations¹⁸ that blood pressures of individuals consuming less than 300 mg per day of elemental Ca^{2+} are 2 to 3 mm Hg higher on average than those of individuals consuming greater than 800 mg per day. The association with blood pressure has been described for dairy product consumption as well as for total calcium intake. This observation is not surprising, because in industrialized societies about 80% of dietary Ca^{2+} is derived from dairy products.¹⁸ Decreased consumption of dairy products because of lactose intolerance¹⁹ likely explains, in part, the lesser intake of Ca^{2+} in the American black population when compared with that in a comparable white population. Despite the generally consistent observation of an inverse relationship between Ca^{2+} and blood pressure, it has been suggested that calcium intake is merely a surrogate for other social, environmental, or dietary factors that influence blood pressure.¹⁸ For example, individuals with lower educational, social, and income status consume less calcium (and dairy products) and perhaps replace milk with alcoholic or caffeinated beverages (or both), which have the potential to elevate blood pressure.^{18,19} According to this notion, the more critical determinant of blood pressure is not Ca^{2+} but an unmeasured or unanalyzed characteristic or behavior, such as poverty or alcohol consumption, that has a direct relationship.^{12,18} Although attempts have been made to adjust for some of these relevant variables, other confounding variables associated with Ca^{2+} or milk consumption cannot be excluded as not being of primary importance.

ABNORMALITIES IN CELLULAR CALCIUM METABOLISM IN HYPERTENSION

Evidence has accumulated that Ca^{2+} metabolism is deranged in a subset of the population with essential hypertension. These alterations in calcium metabolism

may lead to increases in intracellular Ca^{2+} [Ca^{2+}]_i by a variety of mechanisms, as discussed below. Such increases in [Ca^{2+}]_i, in turn, result in increased vascular smooth muscle contraction and thereby increase peripheral vascular resistance.²⁰⁻²² Increased [Ca^{2+}]_i has been reported in platelets,^{23,24} erythrocytes,²⁵⁻²⁸ and adipocytes²⁹ of patients with essential hypertension and in animal models of hypertension, and vascular smooth muscle Ca^{2+} influx and total aortic Ca^{2+} in the SHR have been reported to increase when compared with those in controls.³⁰ The level of renal tubular [Ca^{2+}]_i, however, has been reported to be either unchanged³¹ or reduced in SHRs.³² This increase has been attributed both to intrinsic cellular defects and to systemic defects in Ca^{2+} handling that result in alternations in circulating factors (e.g., natriuretic factor, PTH, 1,25-(OH)₂, or 1,25-(OH)₂-D), which then directly or indirectly effect increases in [Ca^{2+}]_i.

Some subjects with hypertension also exhibit reduced binding of Ca^{2+} to the outer side of erythrocyte membranes when compared with subjects without hypertension.³³ In similar fashion, two animal models of hypertension, the deoxycorticosterone acetate-saline hypertensive rat and the SHR, exhibit reduced membrane Ca^{2+} binding in the aorta and in smaller arteries when compared with controls.³⁴⁻³⁶ This reduction in membrane Ca^{2+} binding may facilitate increases in [Ca^{2+}]_i, just as Ca^{2+} binding to the outer aspect of the membrane may inhibit Ca^{2+} influx by inactivating potential-mediated channels.³⁷

Ca^{2+} binding to the inner membrane surface is also reduced in erythrocytes from human subjects with hypertension^{33,38,39} when compared with those from subjects without hypertension, and a similar abnormality has been identified in plasma membrane preparations from several tissues (i.e., erythrocytes and heart, liver, and neural tissue) in the young SHR when compared with its normotensive control.^{40,41} This suggests that a generalized genetic defect in Ca^{2+} binding to the inner membrane could indirectly cause increases in [Ca^{2+}]_i.^{42,43}

Another mechanism by which [Ca^{2+}]_i may be increased in hypertensive states would be a reduction in Ca^{2+} -ATPase activity. Vincenzi et al.⁴⁴ noted that white subjects with hypertension demonstrated reductions in erythrocyte Ca^{2+} -ATPase activity. Because the level and distribution of calmodulin does not appear to be abnormal in the hypertensive state,⁴⁵ the observed decrease in Ca^{2+} pump activity may reflect impaired calmodulin binding, with a subsequent failure to activate membrane Ca^{2+} -ATPase activity, or an intrinsic defect in the pump itself, leading to reduced calmodulin activation of Ca^{2+} -ATPase. Such a defect in platelet mem-

brane Ca^{2+} -ATPase was observed by one group who reported that calmodulin activation of Ca^{2+} -ATPase was reduced in platelets from patients with hypertension when compared with those in subjects without hypertension.⁴⁶ Erythrocyte Ca^{2+} -ATPase activity in black patients with hypertension has been observed⁴⁷ to be about 30% lower than that in white subjects.⁴⁴ This could explain, in part, the high levels of platelet²⁴ and erythrocyte $[\text{Ca}^{2+}]_i$ ⁴⁷ previously observed in black individuals with hypertension.

Lindner et al.⁴⁸ recently found that incubation of platelets from subjects without hypertension with plasma from patients with hypertension resulted in a substantial (80%) increase in platelet $[\text{Ca}^{2+}]_i$, but incubation of platelets from patients with hypertension with plasma from controls without hypertension resulted in a 33% decrease in platelet $[\text{Ca}^{2+}]_i$. Thus, there is ample evidence to indicate that a circulating factor may be partially responsible for elevations in $[\text{Ca}^{2+}]_i$. It cannot be concluded, however, that a circulating Na^+ - K^+ -ATPase inhibitor is that factor, because the transport activity of sodium has not been measured concomitantly with that of Ca^{2+} in such incubation experiments. Alternatively, it is possible that Ca^{2+} regulatory hormones (PTH or $1,25\text{-(OH)}_2\text{-D}$ [or both]) are directly responsible for such increases in intracellular Ca^{2+} .

ROLE OF CALCIUM REGULATORY HORMONES

The Ca^{2+} regulatory hormones PTH and $1,25\text{-(OH)}_2\text{-D}$ have been observed to be elevated in a subset of the hypertensive population.⁴⁷⁻⁵³ The hypertensive group that tends to have elevated levels of these Ca^{2+} regulatory hormones is the group of individuals with salt sensitivity^{47,53} and low renin levels,^{47,50-53} who tend to have low ionized Ca^{2+} levels⁴⁹⁻⁵³ and increased calcitriol.⁴⁷⁻⁵³ Both PTH and $1,25\text{-(OH)}_2\text{-D}$ may have blood pressure-elevating properties. PTH has the ability to stimulate cellular Ca^{2+} uptake and thereby increase $[\text{Ca}^{2+}]_i$. PTH has been shown to increase Ca^{2+} uptake in a variety of cells, including erythrocytes⁵⁴ and isolated aortic tissue.⁵⁵ A hypertensinogenic effect of PTH is supported by the finding that the pressor response to Ca^{2+} infusion in Sprague-Dawley rats is reduced by parathyroidectomy.⁵⁶ In similar fashion, the hypertensive effect of acute hypercalcemia in human subjects is realized only with intact parathyroid function and is completely abolished by parathyroidectomy,⁵⁷ further suggesting a causative role for PTH in hypertension.

Additional evidence for a hypertensive effect of PTH is the recent reported evidence that increases in serum PTH and urinary cyclic adenosine monophosphate induced by increasing dietary salt intake are accompanied by significant increases in erythrocyte $[\text{Ca}^{2+}]_i$ and blood pressure,⁴⁷ and that blocking the increase in PTH by raising dietary calcium caused both $[\text{Ca}^{2+}]_i$ and blood

pressure to return to baseline levels. PTH may also increase $[\text{Ca}^{2+}]_i$ by a sodium-dependent mechanism, because Acceto and Weder⁵⁸ recently found that PTH inhibits erythrocyte Na^+ - K^+ -ATPase.

It has been suggested that $1,25\text{-(OH)}_2\text{-D}$ also may exert a hypertensive effect.⁵¹⁻⁵³ It has been reported⁵⁹ that nanomolar quantities of the hormone caused a threefold increase in calcium 45 uptake; as with PTH, this $1,25\text{-(OH)}_2\text{-D}$ stimulation of Ca^{2+} uptake was found to exhibit a biphasic response, because maximal stimulation occurred at a 1.0 nmol/L concentration, but no stimulation occurred at 10 nmol/L. In addition, Resnick et al.^{51,52} reported that subjects with hypertension with low renin levels exhibited a hypotensive response to oral Ca^{2+} supplementation in the absence of vitamin D, but that the addition of 0.25 μg $1,25\text{-(OH)}_2\text{-D}$ resulted instead in a pressor response. Thus, available evidence indicates that observed elevations in both PTH and $1,25\text{-(OH)}_2\text{-D}$ in low-renin, salt-sensitive hypertension likely play a role in the pathogenesis of salt-sensitive hypertension via stimulation of Ca^{2+} influx.

DIETARY CALCIUM INTERVENTION STUDIES

Experimental evidence has accumulated over the last decade which suggests that the suboptimal calcium intakes and the abnormalities in Ca^{2+} metabolism previously outlined may play a role in the genesis and maintenance of hypertension, particularly in the low-renin, salt-sensitive state in humans and in animal models of salt-sensitive hypertension. The remaining discussion will deal only with dietary intervention studies in humans.

Over 20 dietary Ca^{2+} intervention studies in humans that use a variety of intervention periods (2 to 48 weeks), sample sizes, and populations have been published⁶⁰⁻⁸² (Table I). Both Ca^{2+} salts (carbonate, citrate, phosphate, gluconate, lactate, and dietary Ca^{2+} [usually dairy foods]) have been used in these intervention studies. Both subjects without hypertension and subjects with hypertension have been studied with variable results. For example, in unblinded studies conducted in Central America, significant reductions in blood pressure in nonpregnant⁶⁰ and pregnant women⁶¹ without hypertension who received supplements of 1 gm daily of elemental Ca^{2+} were observed. In similar fashion, Zemel et al.,⁶² in an unblinded study, observed that an increase in Ca^{2+} intake of approximately 350 to 950 mg daily would significantly reduce salt-induced rises in systolic and diastolic blood pressures in black subjects who had salt sensitivity with or without hypertension. In a randomized, double-blind, placebo-controlled 12-week trial, a 1500 mg/day Ca^{2+} supplementation resulted in a modest but significant lowering of mean arterial blood pressures in both white and black

Table 1. Comparison of calcium intervention trials

Calcium dose	Population	Study design	Response	Reference
1000 mg	Pregnant and nonpregnant women without hypertension	Randomized, double-blind placebo-controlled 22-week duration	Decreased diastolic pressure, 5.6% in women and 9% in men	60,61
600 mg	Blacks without hypertension	Unblinded crossover 2-week periods	Decreased systolic pressure, 4 mm Hg	62
600 mg	Blacks with hypertension	Unblinded; 2-week periods	Decreased systolic pressure, 12 mm Hg; decreased diastolic pressure, 4 mm Hg	26,47
1500 mg	White and black men without hypertension	Randomized double-blind placebo-controlled 12-week duration	Decreased systolic pressure, 3 mm Hg; decreased diastolic pressure, 2 mm Hg	63
1000 mg	Subjects with and without hypertension	Randomized, double-blind placebo-controlled 8-week duration	No effect in subjects without hypertension; decreased systolic pressure of 4 mm Hg and diastolic pressure of 2 mm Hg in subjects with hypertension	64
1000 mg	Subjects with and without hypertension	Single-blind placebo-controlled crossover, 8-day periods	No effect in subjects without hypertension; decreased systolic pressure of 4 mm Hg in subjects with hypertension	65
2000 mg	Subjects with hypertension	Unblinded	Decreased diastolic pressure, 7 mm Hg at 5 days and 5 mm Hg after 1 month	67,68
1000 mg	Young white subjects with hypertension	Randomized placebo-controlled double-blind 12-week duration	Decreased diastolic pressure, 2 to 3 mm Hg	69
1000 mg	Subjects with hypertension	Randomized, double-blind, placebo-controlled crossover, 15-week duration	Decreased systolic pressure, 9 mm Hg	70
2000 mg	Elderly subjects with hypertension	Placebo-controlled crossover, 8-week duration	Decreased systolic pressure, 24 mm Hg; decreased diastolic pressure, 15 mm Hg	71
1500 mg	Women subjects beyond menopause who have hypertension	Randomized, placebo-controlled double-blind 4-year duration	Decreased systolic pressure, 13 mm Hg	72
1000 mg	Elderly women	3-Month duration	Decreased systolic pressure, 5 mm Hg; decreased pressure, 3 mm Hg	73
1000 mg	Subjects with hypertension	Randomized, placebo-controlled double-blind crossover, 8-week duration	No effect	75
800 mg	Subjects with hypertension	Randomized, placebo-controlled double-blind crossover, 2-month duration	No effect	81

men without hypertension.⁶³ In contrast, McCarron and Morris⁶⁴ reported that dietary supplementation with 1 gm of elemental Ca^{2+} daily had no effect on blood pressure in 32 men without hypertension. Luft et al.⁶⁵ also observed no blood pressure lowering effect during 8 days of supplementation with 1000 mg/day of ele-

mental Ca^{2+} in eight individuals without hypertension but a reduction of 4% in systolic blood pressure in subjects with hypertension.

Ca^{2+} interventional trials in hypertensive populations have also yielded variable results. Resnick et al.⁶⁶⁻⁶⁸ conducted very short-term (5-day) and long-term (1-

month) studies and reported that Ca^{2+} supplementation decreased diastolic blood pressure by 7 mm Hg at 5 days and by 5 mm Hg at one month. Resnick et al.⁶⁷ also reported a modest decline (4%) in diastolic blood pressure in subjects with hypertension who received supplementation with 2 gm of elemental Ca^{2+} daily. These investigators noted that their hypertensive population varied in their responses to Ca^{2+} supplementation and identified a subgroup with low serum Ca^{2+} levels and low renin levels who responded to Ca^{2+} supplementation.⁶⁸ McCarron and Morris,⁶⁴ in their randomized placebo-controlled trial, found that subjects with hypertension supplemented with 1 gm of elemental Ca^{2+} over a 2-month period decreased supine systolic blood pressure by 3.8 mm Hg, standing systolic pressure by 5.6 mm Hg, and standing diastolic pressure 2.3 mm Hg, respectively. These investigators noted that a subgroup (47%) of their white subjects with hypertension had reductions in standing systolic pressures of 10 mm Hg or more, indicating that individuals with higher initial blood pressures were more likely to respond. Grobbee and Hofman⁶⁹ observed that white patients with mild hypertension given treatment with an oral supplementation of 1 gm daily of elemental Ca^{2+} had a reduction in diastolic blood pressure at 6 and 12 weeks of 3.1 and 2.4 mm Hg, respectively, when compared with placebo treatment groups. They observed an antihypertensive effect in one third to one half of patients given supplementation with Ca^{2+} . These investigators further noted that their young adults with mild hypertension who had greater than median baseline PTH levels had the greater reduction in diastolic pressure with Ca^{2+} supplementation. Strazzullo et al.⁷⁰ reported that patients with essential hypertension given 1 gm/day Ca^{2+} had a decrease at the end of 15 weeks only in standing systolic pressure when compared with a placebo group. Tabuchi et al.⁷¹ noted a relatively consistent blood pressure-lowering effect of long-term elemental Ca^{2+} supplementation in elderly patients with essential hypertension. Johnson et al.⁷² reported that a daily Ca^{2+} supplement of 1500 mg per day caused a 13 mm Hg reduction in systolic blood pressure in 16 women beyond menopause with hypertension who were observed over a 4-year period of supplementation, but that the control group of eight women given a placebo over the same period of time experienced a 7 mm Hg increase in systolic blood pressure. Gilliland et al.⁷³ also observed a reduction in seated systolic blood pressures of approximately 5 mm Hg and diastolic pressures of approximately 3 mm Hg after 3 months of calcium carbonate (1 gm/day) supplementation in older women (mean age 57 years). Similar reductions in systolic and diastolic blood pressures have been observed in black subjects with salt-sensitivity and hypertension after

Ca^{2+} supplementation with approximately 60 mg of dietary Ca^{2+} .^{26,47,52}

In contrast to the aforementioned positive studies, other investigators have reported no antihypertensive effects of oral Ca^{2+} supplementation.⁷⁵⁻⁸² Indeed, some of these studies have been well controlled^{75-77,82} and some double-blind.^{75,77,79} For example, Zoccali et al.⁷⁵ conducted a double-blind, randomized, crossover trial of Ca^{2+} supplementation that was similar in design to that of McCarron and Morris⁶⁴ in white subjects with mild to moderate hypertension. In 21 subjects who completed a 2-month course of oral Ca^{2+} supplementation (1 gm/day), arterial pressure was not different from that associated with a placebo course. These investigators interpreted their results as indicating that oral Ca^{2+} supplementation does not lower arterial pressure in white middle-aged subjects with mild to moderate essential hypertension.

It should be noted that subjects in this study demonstrated variable blood pressure responses, with approximately 10% displaying an antihypertensive response to oral Ca^{2+} supplementation. Meese et al.⁸¹ also reported variable but no statistical blood pressure response in 26 subjects with hypertension who were given 800 mg/day of elemental Ca^{2+} as either carbonate or citrate over a 2-month course. These negative data underline the fact that many individuals in the essential hypertension population will likely not have blood pressure-lowering responses to Ca^{2+} supplementation. These data also underscore the importance of carefully characterizing the hypertensive populations and their diets before evaluating Ca^{2+} supplementation.

Thus, the results of oral Ca^{2+} studies to date suggests that antihypertensive effects are quite variable and are likely to be seen only in certain subgroups of the general population of subjects with essential hypertension. An accumulating body of evidence suggests that certain characteristics may help predict individuals who will have a blood pressure reduction in response to oral Ca^{2+} supplementation. Epidemiologic data¹⁻¹⁹ suggests that the most potent association of dietary Ca^{2+} with blood pressure occurs with Ca^{2+} intakes considerably below the RDA for dietary Ca^{2+} . Accordingly, minimal antihypertensive effects would be anticipated if oral Ca^{2+} supplementation is given to a population whose Ca^{2+} intake is already at or above the RDA of 800 to 1000 mg of elemental Ca^{2+} per day. Consistent with this notion is the observation by Zemel et al.^{26,47,62,74} that blacks who have been shown in careful dietary recall studies^{1,16,19,83} to ingest, on average, 300 to 500 mg of elemental Ca^{2+} daily have rather consistent antihypertensive responses to supplementary oral Ca^{2+} in amounts sufficient to elevate their daily intake up to the RDA for Ca^{2+} . Observations of good antihypertensive

responses to oral Ca^{2+} supplementation in women beyond menopause^{72,73} and in older men⁷¹ are also consistent with this notion, in consideration of data suggesting that daily Ca^{2+} intake is below the RDA in the elderly.^{1,73,83,84} Unfortunately, most of the Ca^{2+} intervention studies, to date, have not evaluated the pre-supplement Ca^{2+} intake of the study population. Further, epidemiologic data¹⁻¹⁹ suggest that the strongest association between Ca^{2+} intake and blood pressure occurs with Ca^{2+} intakes of less than 600 mg/day and that the association is increasingly diminished with intakes above 800 mg/day. Accordingly, use of Ca^{2+} supplementation with 1 to 2 gm, as done in many of the reported studies, is probably inappropriate.

BLOOD PRESSURE RESPONSE TO ORAL CALCIUM: PREDICTIVE VARIABLES

Although there has been no absolute predictor of a blood pressure response to increased Ca^{2+} intake, several characteristics of responding populations may be important. Resnick et al.⁵¹ reported that patients with essential hypertension who exhibited a profile of low renin, lower ionized Ca^{2+} , and elevated PTH and $1,25\text{-(OH)}_2\text{-D}$ levels displayed a decrease in blood pressure in response to oral Ca^{2+} supplementation, in contrast to those with opposite biochemical profiles. Grobbee and Hofman⁶⁹ also reported that young adults with mild hypertension who had greater than mean PTH levels and lower than mean Ca^{2+} levels were more likely to respond to oral Ca^{2+} supplementation. Zemel et al.,^{26,47,62,74} from the results of their Ca^{2+} supplementation studies, have suggested that subjects with salt sensitivity are more likely to demonstrate blood pressure reductions in response to dietary Ca^{2+} supplementation. According to this notion, the individual with salt sensitivity must be consuming a high-salt diet to demonstrate a blood pressure-reducing effect of oral Ca^{2+} supplementation.^{26,47,62,74,84} These investigators, as well as others,⁷³ have also suggested that an early natriuretic response in the first several days after institution of Ca^{2+} supplementation may be predictive of a long-term antihypertensive response. Finally, some investigators^{63,69} have observed that subjects having a body weight greater than the median are more likely to exhibit blood pressure reductions after oral Ca^{2+} supplementation.

MECHANISMS OF BLOOD PRESSURE RESPONSE TO ORAL CALCIUM MECHANISMS

The mechanisms by which Ca^{2+} supplementation lowers blood pressure in some individuals remain controversial. Possible mechanisms include changes in smooth muscle membrane stabilization or calcium reg-

ulating hormones (or both), natriuresis and associated changes in sodium homeostasis, direct or indirect effects on cell Ca^{2+} metabolism, and depletion of phosphate or magnesium (or both) (Fig. 1, Table II). The work of Holloway and Bohr⁸⁵ suggests a membrane-stabilizing effect of oral Ca^{2+} that paradoxically prevents Ca^{2+} channel transport and thus induces vascular smooth muscle relaxation. This popular theory is obviously a difficult one to examine in human studies. Accordingly, the remainder of this review will survey the evidence garnered in support of other mechanisms that may explain the blood pressure-lowering effects of Ca^{2+} supplementation. These effects may be primarily limited to individuals with salt-sensitivity-related hypertension.* Evidence from these studies suggests that Ca^{2+} supplementation may help correct a subtle salt-induced negative Ca^{2+} balance and associated increases in PTH and $1,25\text{-(OH)}_2\text{-D}$ in the individual with salt sensitivity. Consistent with this notion is the observation by Resnick et al.^{52,53,68} that patients who demonstrated antihypertensive responses to Ca^{2+} supplementation had salt sensitivity, low plasma renin activity, low serum ionized Ca^{2+} , and elevated PTH and $1,25\text{-(OH)}_2\text{-D}$, but those with normal or high renin activity had normal levels of ionized Ca^{2+} and PTH and either did not respond to oral Ca^{2+} or had pressor responses to this intervention. Zemel et al.^{26,47} observed that increasing the salt intake in individuals with salt sensitivity who were consuming a low level of Ca^{2+} (less than 500 mg/day) resulted in a significant increase in blood pressure and in PTH and erythrocyte Ca^{2+} levels. The rise in PTH accompanied and was postulated to be the result of salt-induced calciuresis and an associated decrease in ionized Ca^{2+} . Furthermore, these investigators observed that increasing dietary Ca^{2+} in conjunction with a high salt intake resulted in significant reductions in both serum PTH and erythrocyte Ca^{2+} levels, along with reversal of the salt-induced increase in blood pressure.^{23,41} As previously noted,⁸⁴ this mechanism may be particularly relevant to elderly individuals with hypertension and to other individuals with salt sensitivity, because the age-associated decline in renal function is associated with an age-related rise in the level of circulating PTH.^{86,87} This age-related increase in PTH is especially striking in blacks,⁸⁸ probably because of a greater decline in renal function with age in blacks, particularly blacks with hypertension.

Another putative mechanism for the antihypertensive effect of oral Ca^{2+} supplementation is through the natriuretic properties of this divalent cation. Studies con-

*References 26, 47, 51-53, 62, 67-69, 74, 84.

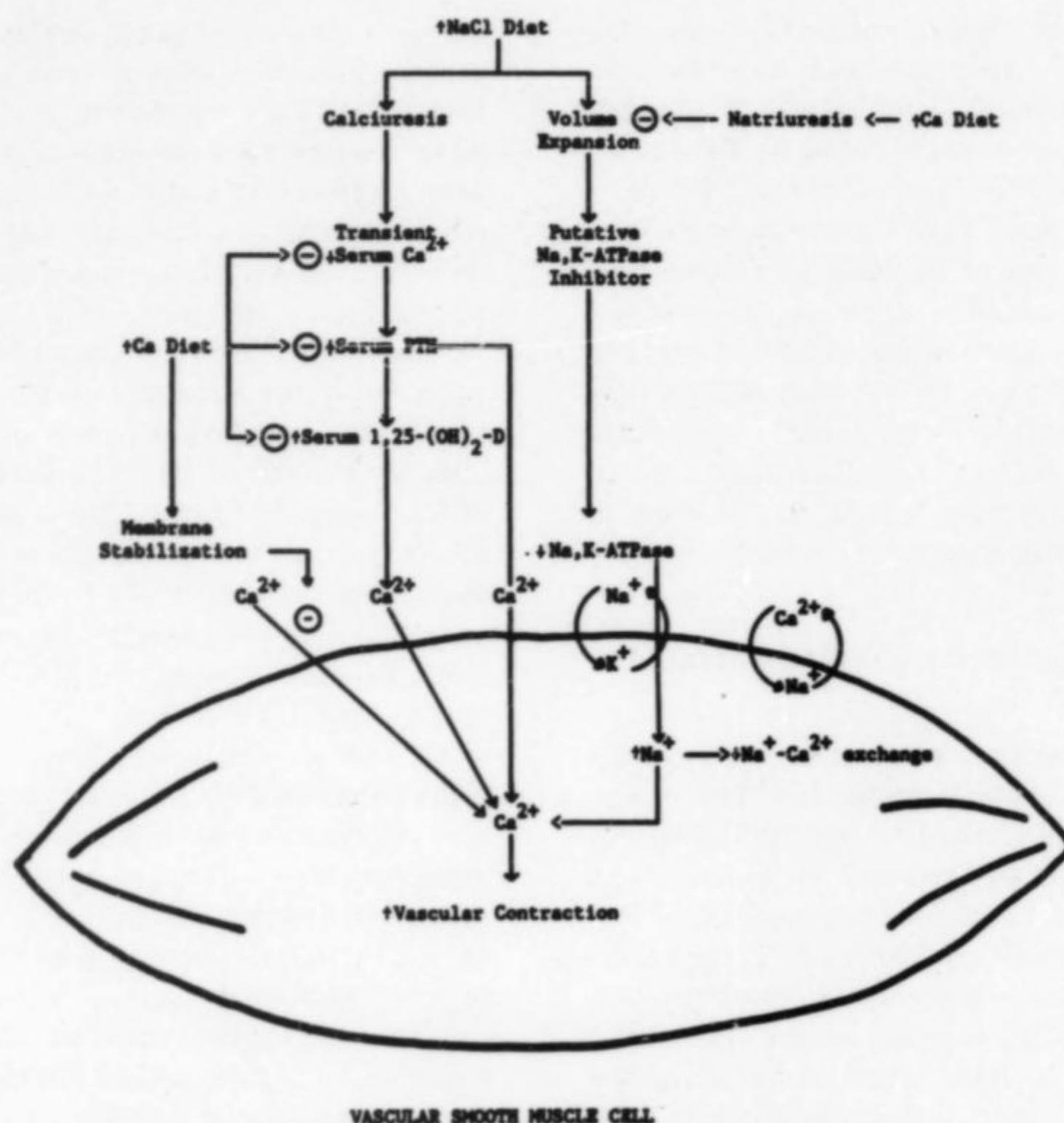


Fig. 1. Effects of dietary calcium on salt-induced changes in vascular resistance.

ducted in SHR that received dietary calcium supplementation support this notion.^{89,90} Zemel et al.^{26,47,74} have observed a significant natriuresis during the first 48 hours after initiating Ca^{2+} supplementation in blacks with salt sensitivity, both with and without hypertension. Ca^{2+} supplementation has also been noted to cause natriuresis in individuals with hypertension and low renin levels⁶⁸ and in women over 40 years of age with hypertension.⁷³ It is interesting that patients with hypertension who have salt sensitivity or low renin levels, or who are elderly, are also the ones who benefit most from diuretic therapy and Ca^{2+} supplementation.

That dietary Ca^{2+} and sodium exert powerful effects on renal excretion of the corresponding cation was first noted by Walser,⁹¹ who reported that an increase of the filtered load of either cation could increase the excretion of the other cation. Others also noted that the urinary excretion of sodium is closely related to that of Ca^{2+} ,⁹² and that these two cations exhibit an integral association in their handling in the proximal convoluted tubule.⁹³ Other recent observations that a significant, sustained calciuresis accompanies salt-induced increases in blood pressure in individuals with hypertension and salt sen-

sitivity who are receiving low Ca^{2+} diets further support such an interrelationship between the renal tubular handling of Ca^{2+} and sodium.^{26,47,62,74} In one study,⁶⁵ however, Ca^{2+} supplementation was not observed to cause natriuresis.

The observed Ca^{2+} -induced natriuresis* may contribute to Ca^{2+} supplementation-related reductions in blood pressure both directly and indirectly. The direct and initial blood pressure-lowering effect in the first several days of Ca^{2+} supplementation may be a consequence of intravascular volume contraction, evidenced both by an increase in plasma renin activity^{26,62} and by significant reductions in total body water and extracellular water levels.⁷⁴ The indirect effects of Ca^{2+} supplementation in ameliorating salt-induced rises in blood pressure may be via prevention of salt-induced and volume-related inhibition of cellular membrane Na-K-ATPase activity.^{47,62,84} Increasing dietary salt ingestion in individuals with salt sensitivity has been shown to be associated with a suppression of erythrocyte mem-

*References 26, 47, 62, 68, 73, 74, 89, 90.

Table II. Possible mechanisms by which dietary calcium supplementation lowers blood pressure in some individuals

Mechanisms	References
Membrane stabilizing effect preventing Ca^{2+} channel transport, leading to decreased vascular tone	85
Activation of membrane Ca-ATPase pump activity with resultant decreased $[\text{Ca}^{2+}]_i$ and vascular tone	44,47
Suppression of PTH-induced increases in $[\text{Ca}^{2+}]_i$ with resultant decrease in vascular tone	47,53,57,84
Suppression of $1,25\text{-(OH)}_2\text{-D}$ -induced increases in $[\text{Ca}^{2+}]_i$ with resultant decrease in vascular tone	51,52,53,84
Calcium-induced natriuresis with associated volume contraction, accompanying intracellular cation changes, and decrease in vascular tone	26,47,62,72,73,86,87
Calcium-induced phosphorus and magnesium depletion with resultant decrease in myocardial contractibility and cardiac output	—

brane Na-K-ATPase activity and a consequent increase in intracellular sodium levels.⁴⁷ If the same phenomenon occurs in vascular smooth muscle tissue, this may result in increased $[\text{Ca}^{2+}]_i$ by virtue of suppressed Na- Ca^{2+} exchange. Ca^{2+} supplementation in the face of high salt intake, by virtue of a natriuretic effect, may effectively decrease the effects of dietary salt in expanding intravascular volume and suppressing Na-K-ATPase and prevent the associated rise in $[\text{Na}]_i$ and $[\text{Ca}^{2+}]_i$ and the accompanying vasoconstriction.

It should be noted that Ca^{2+} supplementation, like diuretic therapy, is associated with only a transient natriuresis^{26,62,74}; however, this is probably adequate to prevent the subtle salt-induced increase in intravascular volume⁷⁴ and the accompanying changes in cellular cation metabolism leading to vasoconstriction in the individuals with salt sensitivity.^{74,84} The observation in most studies that Ca^{2+} supplementation exerts a greater effect on systolic than diastolic blood pressures is also consistent with this proposal. Other proposed mechanisms for the effects of Ca^{2+} supplementation include phosphorus and magnesium depletion and associated deleterious effects on myocardial function.⁸⁴ The associated reduction in cardiac output could also explain the predominant effect of Ca^{2+} supplementation on systolic blood pressures. However, there is virtually no experimental data to support a role for phosphorus and magnesium depletion in the antihypertensive action of Ca^{2+} supplementation. Postulated mechanisms for dietary Ca^{2+} supplementation-related blood pressure reductions are summarized in Table I.

CONCLUSIONS

Dietary Ca^{2+} supplementation to achieve RDA levels of 800 to 1000 mg daily of elemental Ca^{2+} may help correct a subtle chronic deficiency of this divalent cation

and the associated salt-induced alterations in Ca^{2+} homeostasis and blood pressure regulation in persons with salt sensitivity. This appears to be particularly applicable to blacks and the elderly, who typically consume less dietary Ca^{2+} than the rest of the population.⁷⁸ This approach does not require excessive Ca^{2+} supplementation, which has the potential for adverse affects, such as nephrolithiasis, that may be associated with the ingestion of larger quantities of Ca^{2+} (>1500 mg/day). Preferably, this treatment can be accomplished within the framework of an overall dietary modification that would include not only salt restriction, Ca^{2+} supplementation, or both, but also restriction of caloric and saturated fat intake, and an increase in dietary fiber, potassium, and magnesium intake.

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